ORGNZ-07 LAB: Advanced Permission Patterns - Hands-on Exercises

> **Series:** ORGNZ | **Notebook:** 7 of 10 | **Type:** LAB | **Created:** February 2026 | **Last Updated:** 02/19/2026

Overview

This lab notebook contains 3 hands-on exercises extracted from **ORGNZ-07: Advanced Permission Patterns**. Complete the lecture notebook first, then work through these exercises to reinforce the concepts with real DQL queries against your Dynatrace environment.

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Prerequisites

Requirement	Details
Completed	ORGNZ-07: Advanced Permission Patterns (lecture notebook)
Dynatrace Environment	SaaS tenant with Grail enabled
Permissions	`logs.read`, `metrics.read`, `entities.read`, `spans.read`

Exercise 1: Supported Record-Level Conditions

ORGNZ-07: Advanced Permission Patterns

> **Series:** ORGNZ | **Notebook:** 7 of 10 | **Created:** January 2026 | **Last Updated:** 02/09/2026

This notebook covers advanced permission patterns including record-level permissions, field-based access, and combining multiple access control mechanisms for enterprise-scale data governance.

| Requirement | Details |
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1. Record-Level Permissions
2. Record-Level Policy Examples
3. Field-Level Access
4. Combined Permission Patterns
5. Policy Boundaries
6. Enterprise Architecture Patterns
7. Testing Permissions

8. Best Practices

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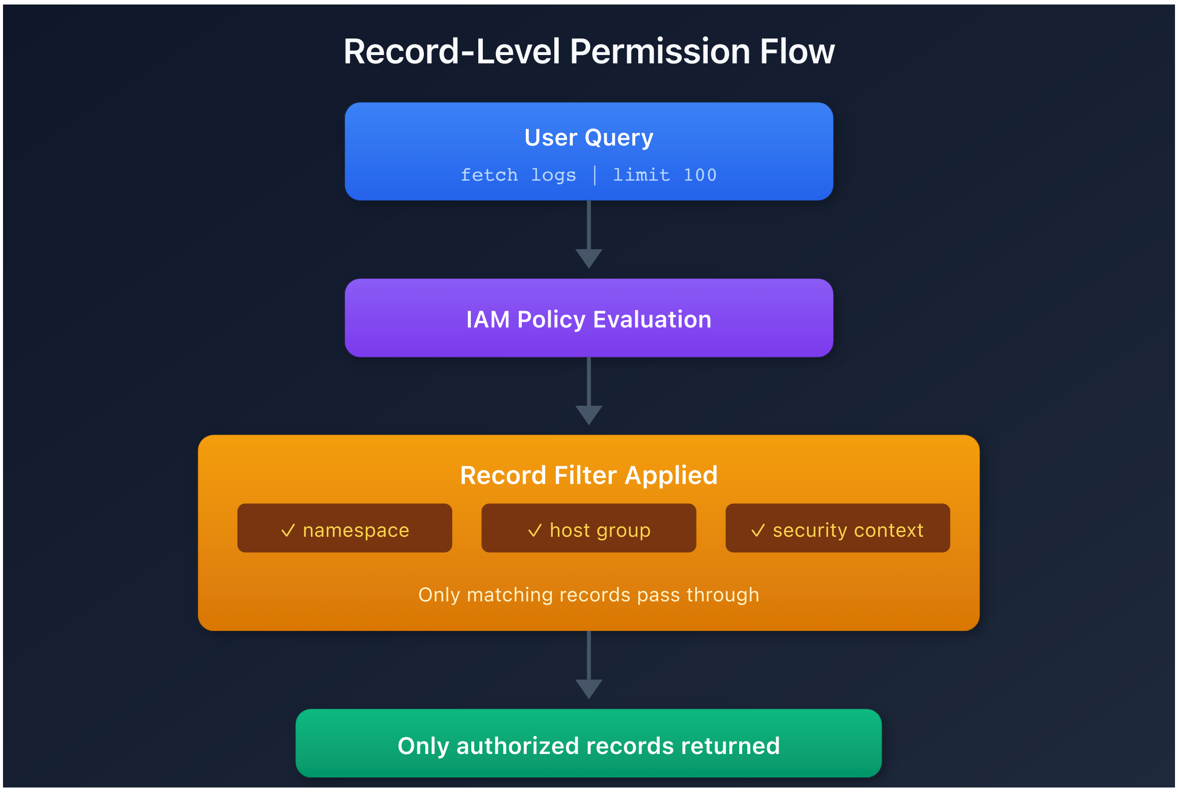
Dynatrace Account	Account-level administrative access
Permissions	IAM policy management permissions
Knowledge	Completed ORGNZ-04 through ORGNZ-06

By the end of this notebook, you will:
- Implement record-level permissions using various attributes
- Understand field-level access restrictions
- Use policy boundaries for reusable condition management
- Combine bucket, record, and field-level policies
- Design enterprise-scale permission architectures

Record-level permissions filter data at query time based on record attributes:



Condition	Description	Example
`storage:k8s.namespace.name`	Kubernetes namespace	`= 'production'`
`storage:k8s.cluster.name`	Kubernetes cluster	`= 'main-cluster'`
`storage:host.name`	Host name	`= 'web-server-01'`
`storage:dt.host_group.id`	Host group	`STARTSWITH 'prod-'`
`storage:aws.account.id`	AWS account	`= '123456789012'`
`storage:gcp.project.id`	GCP project	`= 'my-project'`
`storage:azure.subscription`	Azure subscription	`= 'sub-id'`
`storage:azure.resource.group`	Azure resource group	`= 'my-rg'`
`storage:dt.security_context`	Custom context	`MATCH ('team-*')`

```
{
  "name": "production-namespace-access",
  "description": "Access only to production namespace data",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'default_'; ALLOW storage:logs:read WHERE
storage:k8s.namespace.name = 'production';"
}
```

```
{
  "name": "web-tier-access",
  "description": "Access to web tier hosts only",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'default_'; ALLOW storage:logs:read, storage:metrics:read WHERE
storage:dt.host_group.id STARTSWITH 'web-';"
}
```

```
{
  "name": "prod-web-tier-access",
  "description": "Production web tier only",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'default_'; ALLOW storage:logs:read WHERE
storage:k8s.namespace.name = 'production' AND storage:dt.host_group.id
STARTSWITH 'web-';"
}
```

```
{
  "name": "aws-team-account-access",
  "description": "Access to specific AWS account",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'default_'; ALLOW storage:logs:read, storage:metrics:read,
storage:spans:read WHERE storage:aws.account.id = '123456789012';"
}
```

Control access to specific fields within records:

Use Case	Implementation
Hide sensitive fields	Deny access to specific fields
Mask PII	Field-level policies
Compliance requirements	Restrict access to audit fields

```
{
  "name": "restricted-field-access",
  "description": "Hide sensitive fields from general users",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'default_'; ALLOW storage:logs:read; DENY
storage:logs:read:user.email, storage:logs:read:user.ip_address;"
}
```

Combine bucket and record-level permissions:

```
Layer 1: Bucket access
  ALLOW storage:buckets:read WHERE bucket-name IN ('team_logs',
'shared_logs')

Layer 2: Record filtering
  ALLOW storage:logs:read WHERE k8s.namespace.name = 'team-namespace'

Layer 3: Field masking (optional)
  DENY storage:logs:read:sensitive_field
```

Multiple teams share a bucket with security context isolation:

```
// Team A policy
{
  "name": "team-a-shared-bucket",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name =
'shared_logs'; ALLOW storage:logs:read WHERE storage:dt.security_context
MATCH ('team-a*');"
}

// Team B policy
{
  "name": "team-b-shared-bucket",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name =
'shared_logs'; ALLOW storage:logs:read WHERE storage:dt.security_context
MATCH ('team-b*');"
}
```

```
// Production access (restricted)
{
  "name": "production-access",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'prod_'; ALLOW storage:logs:read WHERE storage:dt.security_context
= 'env:production';"
}

// Non-production access (broader)
{
  "name": "non-production-access",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'default_'; ALLOW storage:logs:read WHERE
storage:dt.security_context MATCH ('env:dev*', 'env:staging*', 'env:qa*');"
}
```

Policy boundaries decouple the "what" (policy) from the "where" (conditions), enabling reusable condition sets that can be applied across multiple policies.

Aspect	Description
Purpose	Bundle conditions for reuse across multiple policies
Scope	Record-level and resource-level restrictions
Relationship	Always used together with a policy — boundaries alone don't restrict anything
Limit	Maximum 10 restrictions per boundary

While **policies** define *which features and data* users can access, **boundaries** define *where* users can access them:

```
Policy: ALLOW storage:logs:read, storage:metrics:read, storage:spans:read
Boundary: storage:k8s.namespace.name = "production"
          storage:dt.host_group.id STARTSWITH "prod-"
```

When assigned together, the user gets read access to logs, metrics, and spans — but only for production namespace data on production host groups.

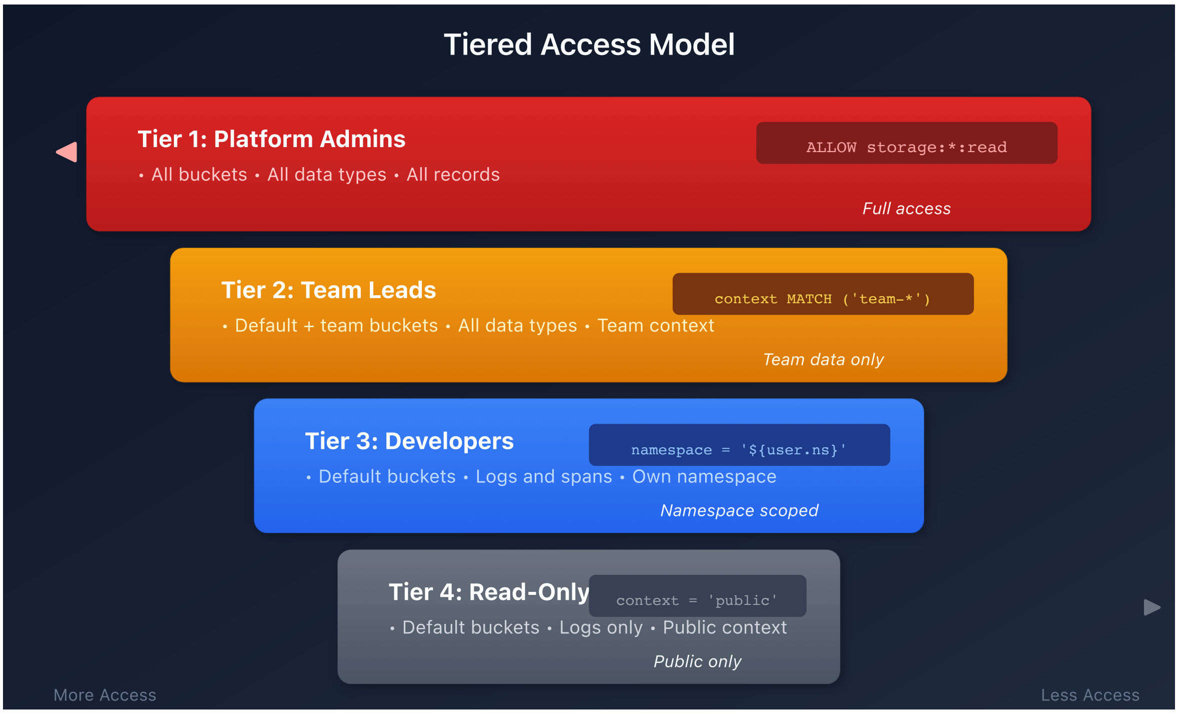
Rule	Detail
No AND operator	Each line is one condition (implicitly AND-combined)
No logical operators	For complex logic, use policy templating
Reusable	Same boundary can be applied to multiple policies
Max 10 restrictions	Create additional boundaries if more are needed

1. Go to **Account Management** > **Identity & access management** > **Policy management**
2. Select the **Boundaries** tab
3. Select **Create boundary**
4. Enter boundary name and conditions (one per line)
5. Save and assign to group policies

```
# EU Region Boundary
storage:bucket-name STARTSWITH "eu_"
storage:azure.subscription = "eu-subscription-id"
```

This boundary can then be applied alongside any policy (logs access, metrics access, etc.) to restrict all data access to the EU region.

> **Tip:** Use boundaries when you have the same set of conditions (e.g., "production only" or "EU region only") that need to apply to multiple policies. This avoids duplicating conditions across every policy definition.



Region	Buckets	Context	Policy
EU	eu_*	region:eu	bucket-name STARTSWITH 'eu_' AND security_context MATCH ('region:eu*')
US	us_*	region:us	bucket-name STARTSWITH 'us_' AND security_context MATCH ('region:us*')

> **Tip:** Use policy boundaries to define regional restrictions once, then apply the same boundary to all team policies within that region.

```
// Test namespace-based access
fetch logs, from:-1h
| filter isNotNull(k8s.namespace.name)
| summarize count = count(), by:{k8s.namespace.name}
| sort count desc
```

Exercise 2: Supported Record-Level Conditions

```
// Test host group access
fetch logs, from:-1h
| filter isNotNull(dt.host_group.id)
| summarize count = count(), by:{dt.host_group.id}
| sort count desc
```

Exercise 3: Supported Record-Level Conditions

```
// Verify accessible data distribution
fetch logs, from:-1h
| summarize
    total = count(),
    withSecurityContext = countIf(isNotNull(dt.security_context)),
    withNamespace = countIf(isNotNull(k8s.namespace.name)),
    withHostGroup = countIf(isNotNull(dt.host_group.id))
```

Lab Summary

You have completed 3 hands-on exercises for **ORGNZ-07: Advanced Permission Patterns**.

Exercises Completed

- [ ] Exercise 1: Supported Record-Level Conditions
- [ ] Exercise 2: Supported Record-Level Conditions
- [ ] Exercise 3: Supported Record-Level Conditions

Next Steps

Continue with **ORGNZ-08** for the next notebook.

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